Chapter 3
Functions and Its Types

Real-Life Story (Why Functions?)

If you travel at a speed of v km/h for t hours, your distance d depends on t:

$$d(t) = v \cdot t.$$

Here, the output (distance) depends on the input (time). Such input-output rules are called **functions**. Functions are everywhere: physics, economics, biology, and computer science.

3.1 Function

Overview

A **function** is a relation that assigns to each input exactly one output. We will learn important types: injective, surjective, bijective, constant, identity, polynomial, rational, even/odd, monotonic, bounded, and floor functions.

3.1.1 Definition

Let A and B be non-empty sets. A **function** f from A to B assigns to each $x \in A$ exactly one $y \in B$:

$$f : A \to B, \quad y = f(x).$$

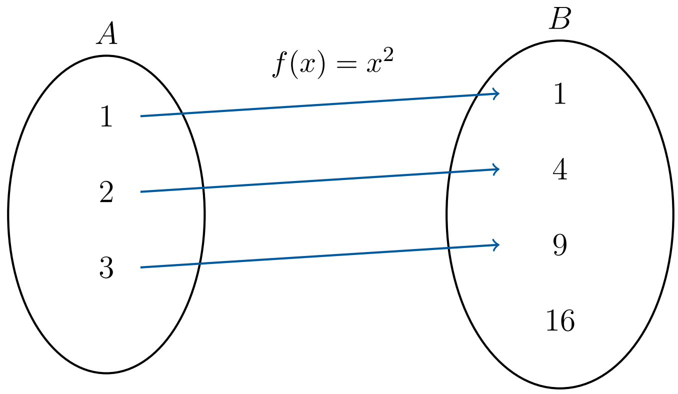
Important Terms

- A is the **domain**
- B is the **codomain**
- The set $\{f(x) : x \in A\}$ is the **range (image)**

Remarks and Notation

- Each input has exactly one output.
- Notation: $f(x)$, $y = f(x)$, $f : A \to B$.
- Example: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, Domain $= \mathbb{R}$, Range $= [0, \infty)$.

Visual Understanding: Mapping Diagram



Domain, Codomain, and Range

Intuition (Why it Matters)
Every function has inputs and outputs. The **domain** tells us what inputs are allowed, the **codomain** defines the possible outputs, and the **range** tells us what outputs actually appear. Understanding these helps us visualize functions and their behavior.

1. Domain: The set of all possible input values for which the function is defined.

$$\text{Domain of } f = \{x \mid f(x) \text{ is defined}\}$$

Example: For $f(x) = \sqrt{x}$, only non-negative numbers are allowed:

$$\text{Domain } = [0, \infty)$$

2. Codomain: The set of values that the function **could possibly produce**, as defined by the function's target set.
Example: For $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$, the codomain is all real numbers:

$$\text{Codomain } = \mathbb{R}$$

3. Range: The set of **actual outputs** the function produces from its domain.
Example: For $f(x) = x^2$ with domain $\mathbb{R}$, the output is always non-negative:

$$\text{Range } = [0, \infty)$$

Term	**Meaning**	**Example** $f(x) = x^2$
Domain	Allowed inputs	$\mathbb{R}$
Codomain	Target outputs	$\mathbb{R}$
Range	Actual outputs	$[0, \infty)$

Tip:

- Domain $\rightarrow$ what you can put in
- Codomain $\rightarrow$ what could come out
- Range $\rightarrow$ what actually comes out

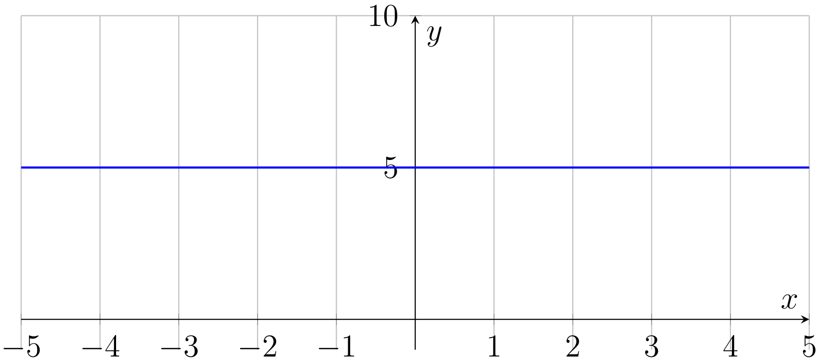
3.1.2 Types of Functions

Constant Function

Definition: A constant function always outputs the same value regardless of the input.

$$f(x) = c, \quad \forall x \in D$$

No matter what x you choose from the domain, $f(x)$ remains c.
Example: $f(x) = 5$
Graph:

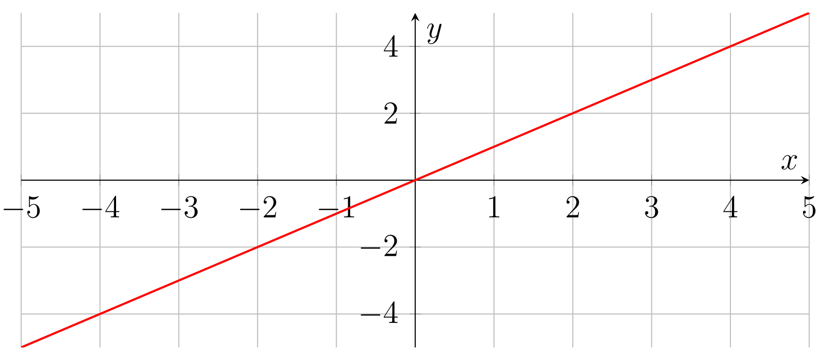


Identity Function

Definition: Identity function maps each input to itself:

$$f(x) = x, \quad \forall x \in D$$

It represents "no change" in the input.
Example: $f(x) = x$
Graph:



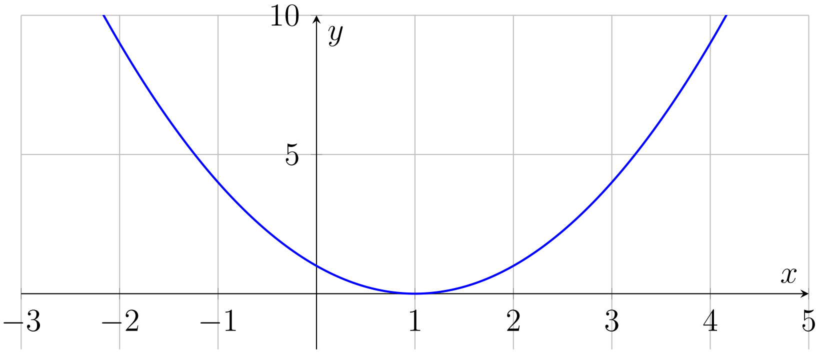
Polynomial Function

Definition: Polynomial functions are sums of powers of x with real coefficients:

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

These functions produce smooth curves and are fundamental in modeling growth or curvature.

Example: $p(x) = x^2 - 2x + 1$
Graph:



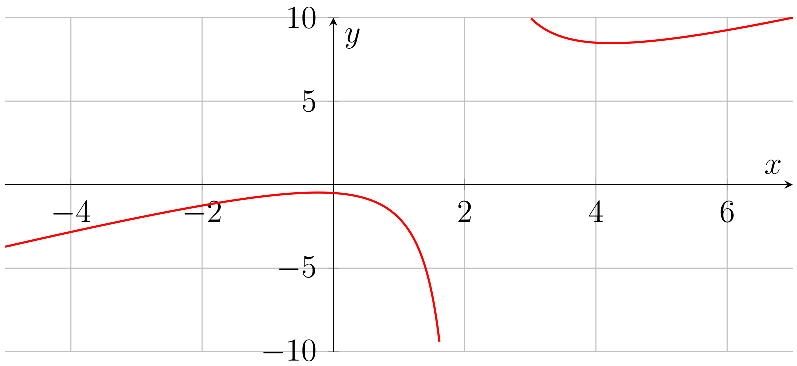
Rational Function

Definition: A rational function is the ratio of two polynomials:

$$r(x) = \frac{p(x)}{q(x)}, \quad q(x) \neq 0$$

It can model rates and hyperbolic behavior; undefined where $q(x) = 0$.

Example: $r(x) = \frac{x^2+1}{x-2}$
Graph:

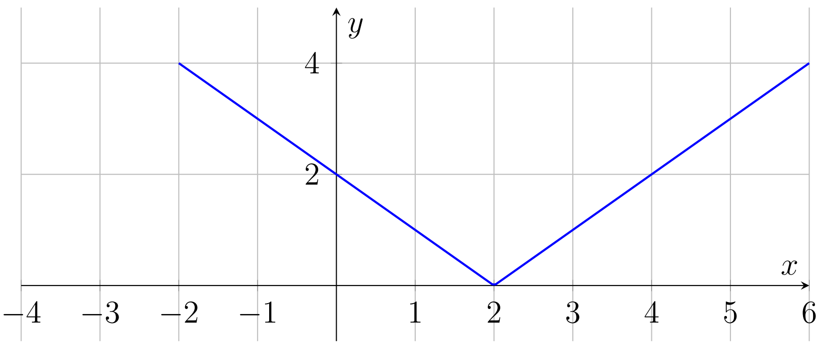


Modulus (Absolute Value) Function

Definition: The modulus or absolute value function gives the distance of a number from zero:

$$f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Example: $f(x) = |x - 2|$
Graph:



Tip:

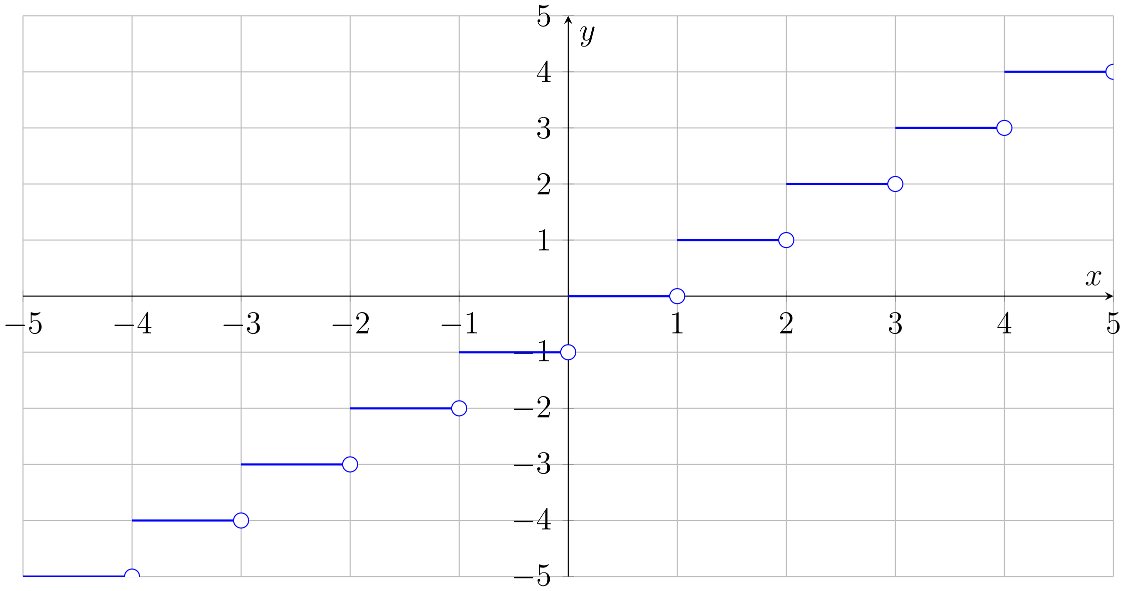
- Graph is V-shaped
- Always non-negative: $|x| \geq 0$
- Symmetric about the y-axis if $f(x) = |x|$

Greatest Integer (Floor) Function

Definition: The greatest integer function, denoted by $\lfloor x \rfloor$, gives the largest integer less than or equal to x:

$$f(x) = \lfloor x \rfloor$$

Example: $f(2.7) = 2, \quad f(-1.3) = -2$
Graph:



Even and Odd Functions

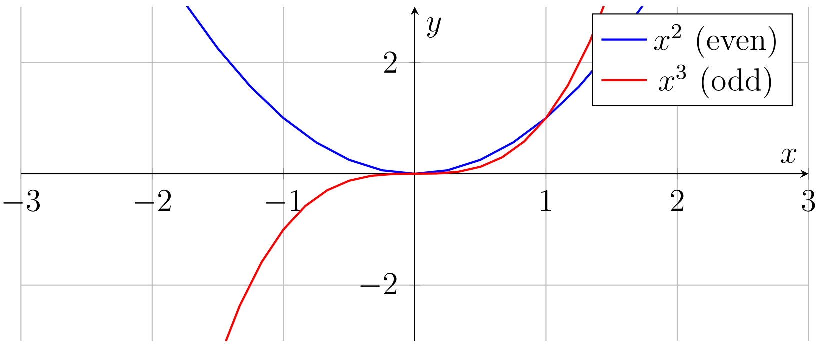
Definition: Symmetry defines even and odd functions:

$$\text{Even: } f(-x) = f(x), \quad \text{Odd: } f(-x) = -f(x)$$

Even functions are symmetric about the y-axis; odd functions are symmetric about the origin.

Example: $f(x) = x^2$ (even), $g(x) = x^3$ (odd)

Graph:



Injective (One-to-One) Function

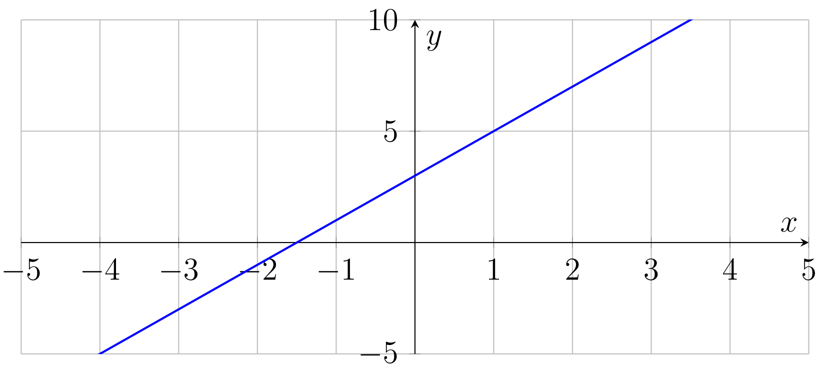
Definition: A function is injective if each output corresponds to exactly one input:

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

Distinct inputs always produce distinct outputs.

Example: $f(x) = 2x + 3$

Graph:



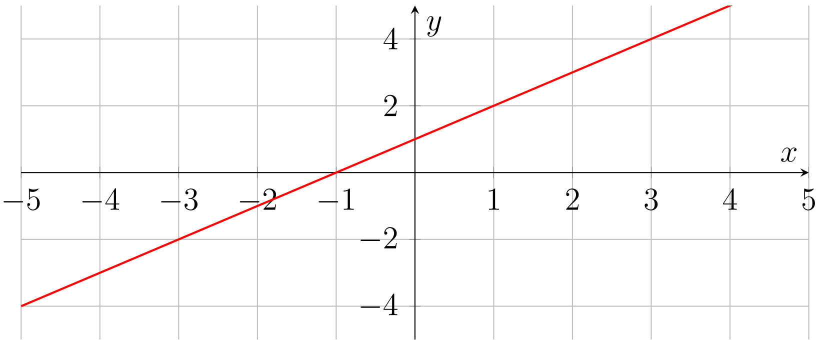
Surjective (Onto) Function

Definition: A function is surjective if every element of the codomain is the image of some element from the domain:

$$\text{Range} = \text{Codomain}$$

Example: $f(x) = x + 1$

Graph:

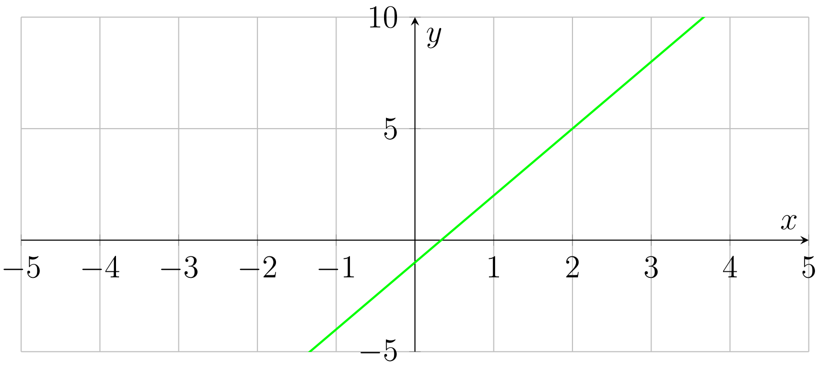


Bijective Function

Definition: A function is bijective if it is both injective and surjective. Each element of the domain maps to a unique element of the codomain, and every element of the codomain has a pre-image. Only bijective functions have inverses.

Example: $f(x) = 3x - 1$, $f^{-1}(y) = \frac{y+1}{3}$

Graph:



Mistake Alert

- Injective $\not\Rightarrow$ Surjective
- Bijective = One-to-One + Onto
- Only bijective functions have inverse functions

Solved Examples:

Example 1: Find the domain of the function:

$$f(x) = \frac{x^2 + 3x + 2}{x^2 - 5x + 6}$$

Solution: The function is undefined when the denominator is 0:

$$x^2 - 5x + 6 = 0 \implies (x-2)(x-3) = 0 \implies x = 2, 3$$

Hence, the domain is:

$$\text{Domain } = \mathbb{R} \setminus \{2, 3\}$$

Example 2: Find the domain and range of the function:

$$f(x) = \sqrt{x + 4}$$

Solution: The expression under the square root must be non-negative:

$$x + 4 \geq 0 \implies x \geq -4$$

Domain: $[-4, \infty)$

Range: $\sqrt{x + 4} \geq 0 \implies [0, \infty)$

Example 3: Find the domain and range of the function:

$$f(x) = |x - 2|$$

Solution: Absolute value is defined for all real numbers:

$$\text{Domain: } \mathbb{R}$$

Range: $|x - 2| \geq 0 \implies [0, \infty)$

Example 4: Let

$$f(x) = \frac{x^2}{1 + x^2}, \quad x \in \mathbb{R}$$

Find the range of f.

Solution: Since $x^2 \geq 0$, we have:

$$0 \leq \frac{x^2}{1 + x^2} < 1$$

Hence, the range is:

$$[0, 1)$$

Example 5: Let $f(x) = x + 2$, $g(x) = 3x - 1$. Find $f + g$, $f - g$, and $\frac{f}{g}$.

Solution:

$$(f + g)(x) = f(x) + g(x) = (x + 2) + (3x - 1) = 4x + 1$$

$$(f - g)(x) = f(x) - g(x) = (x + 2) - (3x - 1) = -2x + 3$$

$$\frac{f}{g}(x) = \frac{f(x)}{g(x)} = \frac{x + 2}{3x - 1}, \quad x \neq \frac{1}{3}$$

Trigonometric Functions

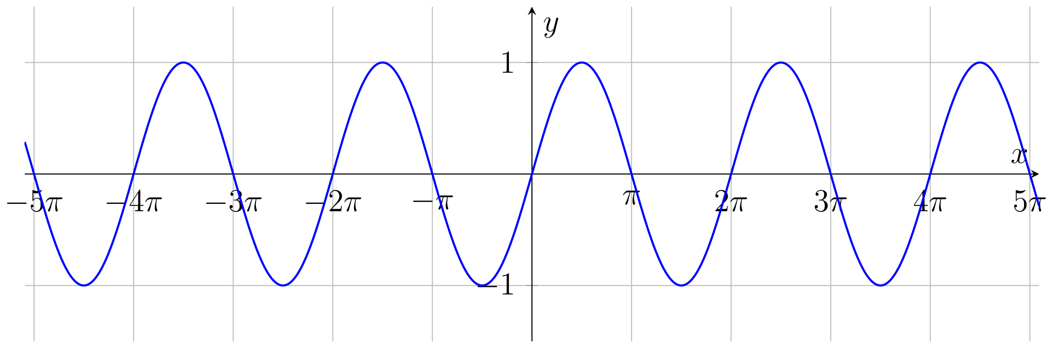
1. Sine Function

Definition: $\sin x$ is the y-coordinate of a point on the unit circle corresponding to angle x radians.

Key Properties of $\sin x$:

- Domain: $(-\infty, \infty)$
- Range: $[-1, 1]$
- Period: 2π
- Symmetry: odd function $(\sin(-x) = -\sin x)$
- $\sin 0 = 0, \sin \frac{\pi}{2} = 1, \sin \pi = 0, \sin \frac{3\pi}{2} = -1$

Graph:



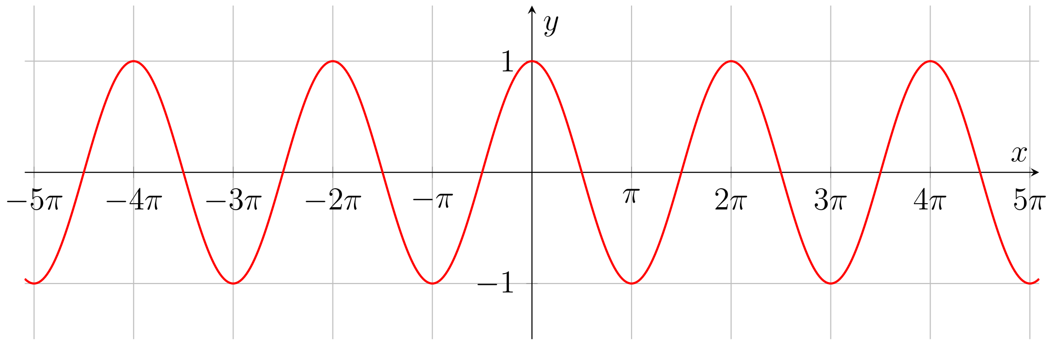
2. Cosine Function

Definition: $\cos x$ is the x-coordinate of a point on the unit circle corresponding to angle x radians.

Key Properties of $\cos x$:

- Domain: $(-\infty, \infty)$
- Range: $[-1, 1]$
- Period: 2π
- Symmetry: even function $(\cos(-x) = \cos x)$
- $\cos 0 = 1, \cos \frac{\pi}{2} = 0, \cos \pi = -1, \cos \frac{3\pi}{2} = 0$

Graph:



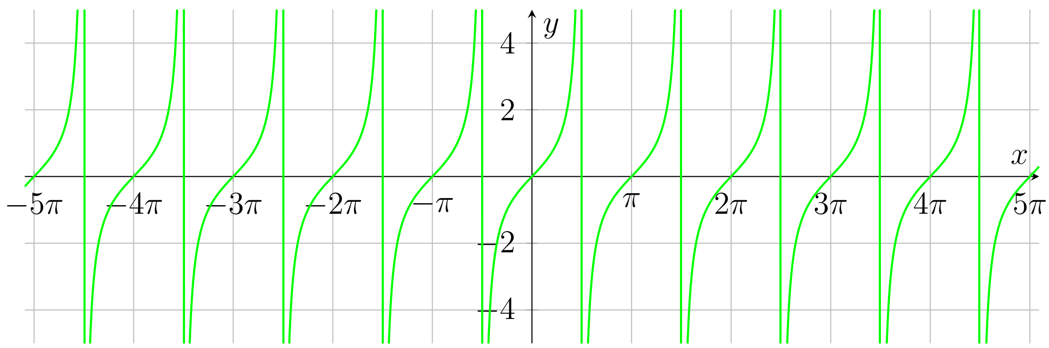
3. Tangent Function

Definition: $\tan x = \frac{\sin x}{\cos x}$, undefined at $x = \frac{\pi}{2} + n\pi$.

Key Properties of $\tan x$:

- Domain: $x \neq \frac{\pi}{2} + n\pi$, $n \in \mathbb{Z}$
- Range: $(-\infty, \infty)$
- Period: π
- Symmetry: odd function
- Vertical asymptotes at $x = \frac{\pi}{2} + n\pi$

Graph:



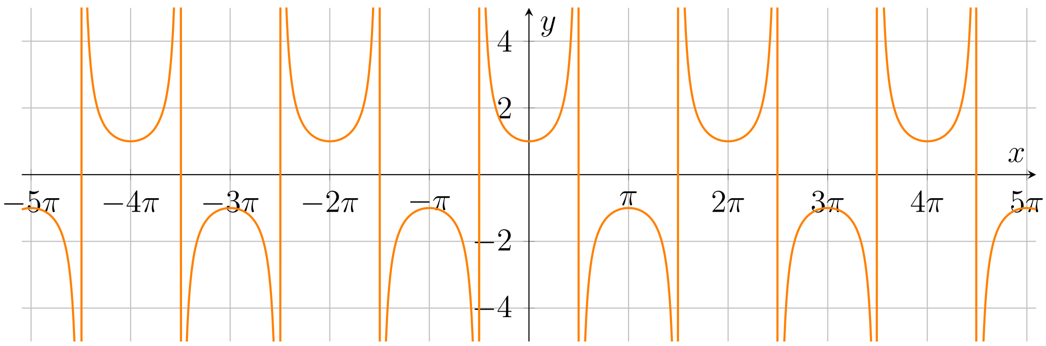
4. Secant Function

Definition: $\sec x = \frac{1}{\cos x}$, undefined at $x = \frac{\pi}{2} + n\pi$.

Key Properties of $\sec x$**:**

- Domain: $x \neq \frac{\pi}{2} + n\pi$, $n \in \mathbb{Z}$
- Range: $(-\infty, -1] \cup [1, \infty)$
- Period: 2π
- Symmetry: even function
- Vertical asymptotes at $x = \frac{\pi}{2} + n\pi$

Graph:



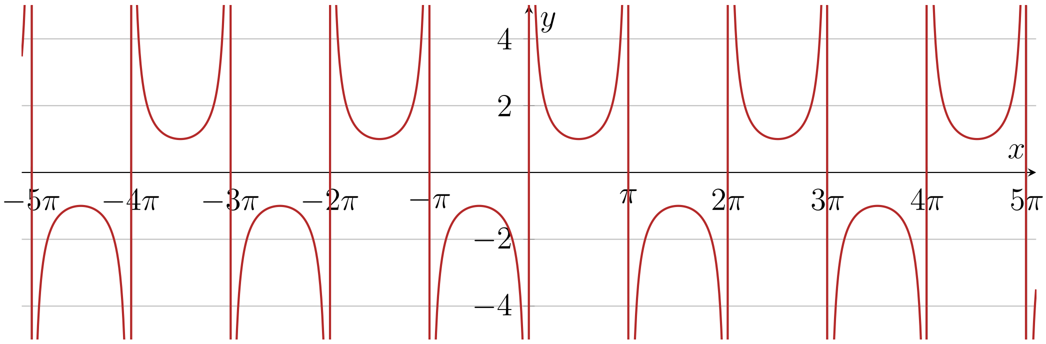
5. Cosecant Function

Definition: $\csc x = \frac{1}{\sin x}$, undefined at $x = n\pi$.

Key Properties of $\csc x$**:**

- Domain: $x \neq n\pi$, $n \in \mathbb{Z}$
- Range: $(-\infty, -1] \cup [1, \infty)$
- Period: 2π
- Symmetry: odd function
- Vertical asymptotes at $x = n\pi$

Graph:



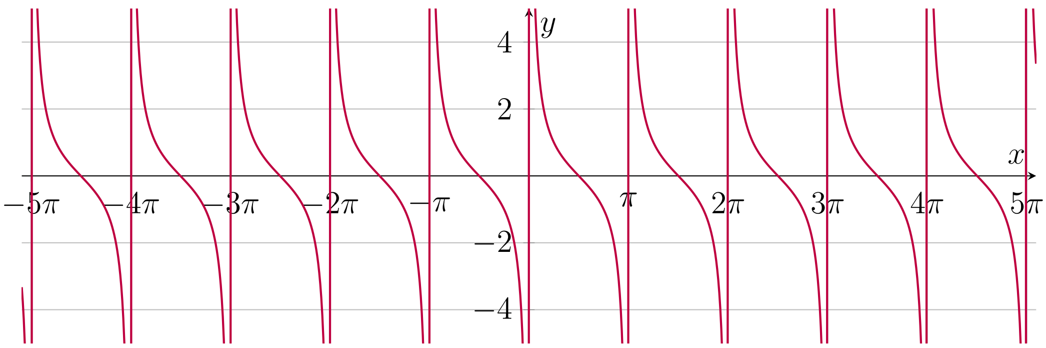
6. Cotangent Function

Definition: $\cot x = \frac{\cos x}{\sin x}$, undefined at $x = n\pi$.

Key Properties of $\cot x$**:**

- Domain: $x \neq n\pi$, $n \in \mathbb{Z}$
- Range: $(-\infty, \infty)$
- Period: π
- Symmetry: odd function
- Vertical asymptotes at $x = n\pi$

Graph:



Periodic Functions

Definition

A function $f(x)$ is called **periodic** if there exists a positive number T such that:

$$f(x + T) = f(x) \quad \forall x \in \text{Domain of } f$$

The smallest such positive number T is called the **fundamental period** of $f(x)$.

Key Points:

- $T > 0$ is the period
- $f(x+T) = f(x)$ holds for all x in the domain
- Functions with period T repeat their values over intervals of length T
- Examples: $\sin x$, $\cos x$, $\tan x$, etc.

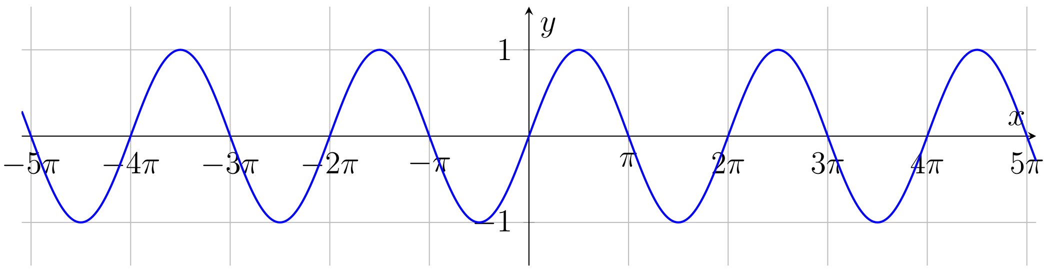
Example 1: Sine Function

Function: $f(x) = \sin x$

Properties:

- Domain: $(-\infty, \infty)$
- Range: $[-1, 1]$
- Period: 2π

Graph:



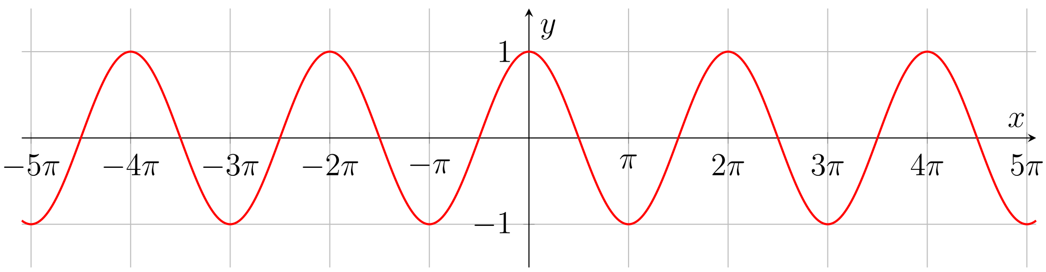
Example 2: Cosine Function

Function: $f(x) = \cos x$

Properties:

- Domain: $(-\infty, \infty)$
- Range: $[-1, 1]$
- Period: 2π

Graph:

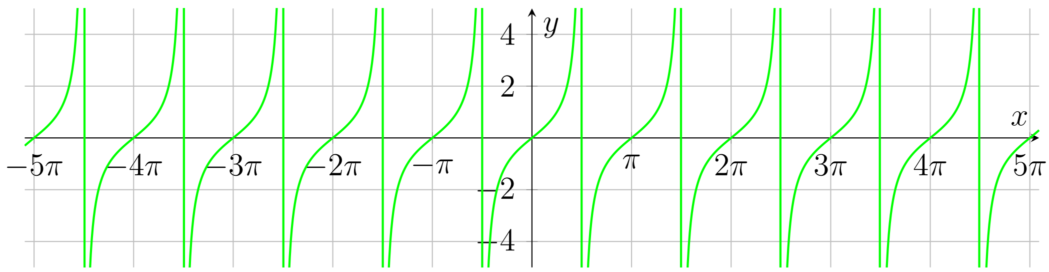


Example 3: Tangent Function

Function: $f(x) = \tan x$
Properties:

- Domain: $x \neq \frac{\pi}{2} + n\pi$, $n \in \mathbb{Z}$
- Range: $(-\infty, \infty)$
- Period: π
- Vertical asymptotes at $x = \frac{\pi}{2} + n\pi$

Graph:



Other Periodic Functions

- $\cot x$, period π
- $\sec x$, period 2π
- $\csc x$, period 2π
- Any function $f(x) = \sin(nx)$ or $f(x) = \cos(nx)$ is periodic with period $\frac{2\pi}{n}$
- Example of non-trigonometric periodic function: $f(x) = |x|$ repeated modulo T (like a sawtooth wave)

Tip: To check if a function $f(x)$ is periodic, try to find $T > 0$ such that $f(x+T) = f(x)$ for all x in its domain.

One-to-One, Onto, and Bijective Functions

Definition

- A function $f : A \to B$ is **injective (one-to-one)** if
$$f(x_1) = f(x_2) \implies x_1 = x_2$$
- A function $f : A \to B$ is **surjective (onto)** if
$$\forall y \in B, \exists x \in A \text{ such that } f(x) = y$$
- A function is **bijective** if it is both injective and surjective. Only bijective functions have inverses $f^{-1} : B \to A$.

Statement

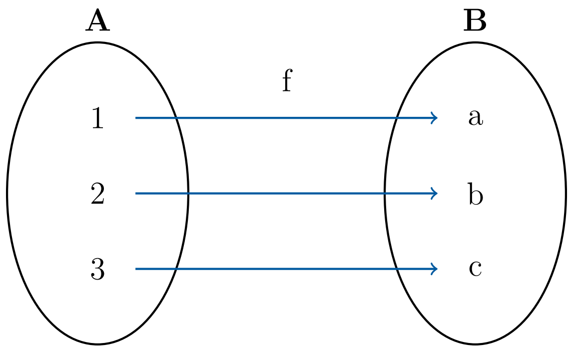
If there exists a one-to-one and onto function between two sets A and B, then:

1. Every element of B has exactly one pre-image in A.
2. The sets A and B have the **same cardinality**.
3. A bijective function has a well-defined inverse $f^{-1} : B \to A$.

Example

Let $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$ and define

$$f(1) = a, \quad f(2) = b, \quad f(3) = c$$



Cardinality of Sets: $\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}$

1. Natural Numbers and Whole Numbers

Sets:

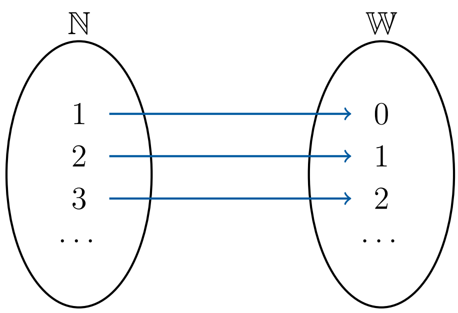
$$\mathbb{N} = \{1, 2, 3, \dots\}, \quad \mathbb{W} = \{0, 1, 2, 3, \dots\}$$

Claim: $\mathbb{N}$ and $\mathbb{W}$ have the same cardinality.
Proof: Define the function:

$$f : \mathbb{N} \to \mathbb{W}, \quad f(n) = n - 1$$

- Injective: $f(n_1) = f(n_2) \implies n_1 = n_2$
- Surjective: $\forall w \in \mathbb{W}, n = w + 1 \in \mathbb{N}$ so $f(n) = w$

Hence, f is bijective and $|\mathbb{N}| = |\mathbb{W}|$.



2. Natural Numbers and Integers

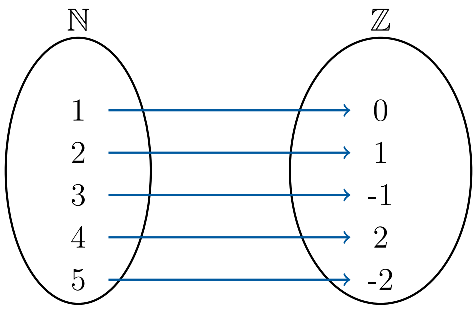
Sets:

$$\mathbb{N} = \{1, 2, 3, \dots\}, \quad \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

Claim: $|\mathbb{N}| = |\mathbb{Z}|$
Bijection: $f : \mathbb{N} \to \mathbb{Z}$ defined by

$$f(n) = \begin{cases} \frac{n}{2}, & n \text{ even} \\ -\frac{n-1}{2}, & n \text{ odd} \end{cases}$$

Example: $f(1) = 0, f(2) = 1, f(3) = -1, f(4) = 2, f(5) = -2, \dots$



3. Natural Numbers and Rational Numbers

Claim: $|\mathbb{N}| = |\mathbb{Q}|$
Proof: Arrange positive rationals $\mathbb{Q}^+$ in a 2D grid:

$$\begin{array}{cccc} \frac{1}{1} & \frac{1}{2} & \frac{1}{3} & \cdots \\ \frac{2}{1} & \frac{2}{2} & \frac{2}{3} & \cdots \\ \frac{3}{1} & \frac{3}{2} & \frac{3}{3} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{array}$$

Enumerate diagonally and skip duplicates to get a sequence:

$$1/1, 1/2, 2/1, 1/3, 3/1, 2/3, 3/2, \dots$$

Extend to negative rationals and 0:

$$0, 1/1, -1/1, 1/2, -1/2, 2/1, -2/1, \dots$$

- This defines a bijection $f : \mathbb{N} \to \mathbb{Q}$. - Hence, $|\mathbb{N}| = |\mathbb{Q}|$.

Diagonal enumeration of positive rationals:

$$\begin{array}{ccc} \frac{1}{1} & \frac{1}{2} & \frac{1}{3} \\ \frac{2}{1} & \frac{2}{2} & \frac{2}{3} \\ \frac{3}{1} & \frac{3}{2} & \frac{3}{3} \end{array}$$

Traverse diagonally and remove duplicates $\to$ enumerates $\mathbb{Q}^+$

Cardinality of Intervals $[a, b]$ and $[c, d]$

Statement

Let $[a, b]$ and $[c, d]$ be two closed intervals of real numbers with $a < b$ and $c < d$. Then there exists a **bijective function** $f : [a, b] \to [c, d]$. Hence, $[a, b]$ and $[c, d]$ have the same cardinality.

Proof

Define a linear function:

$$f(x) = c + \frac{d - c}{b - a}(x - a)$$

Check properties:

- **Injective:** If $f(x_1) = f(x_2)$, then

$$c + \frac{d - c}{b - a}(x_1 - a) = c + \frac{d - c}{b - a}(x_2 - a) \implies x_1 = x_2$$

- **Surjective:** For any $y \in [c, d]$, choose

$$x = a + \frac{b - a}{d - c}(y - c) \in [a, b] \implies f(x) = y$$

Hence, f is bijective and $[a, b]$ and $[c, d]$ have the same cardinality.

Example: Intervals $[1, 2]$ and $[1, 10]$

Sets:

$$[a, b] = [1, 2], \quad [c, d] = [1, 10]$$

Bijection: Define a linear function mapping $[1, 2] \to [1, 10]$:

$$f(x) = 1 + \frac{10 - 1}{2 - 1}(x - 1) = 1 + 9(x - 1) = 9x - 8$$

Check:

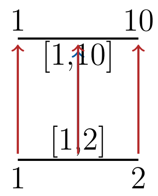
$$f(1) = 9 \cdot 1 - 8 = 1, \quad f(2) = 9 \cdot 2 - 8 = 10$$

Properties:

- **Injective:** $f(x_1) = f(x_2) \implies 9x_1 - 8 = 9x_2 - 8 \implies x_1 = x_2$
- **Surjective:** $\forall y \in [1, 10], x = \frac{y+8}{9} \in [1, 2]$ so $f(x) = y$

Hence, f is **bijective**, and $[1, 2]$ and $[1, 10]$ have the same cardinality.

Graphical Illustration



Inverse Function:

$$f^{-1}(y) = \frac{y+8}{9}, \quad y \in [1, 10]$$

This shows that $[1, 2]$ and $[1, 10]$ are in **one-to-one correspondence**.